
notification-service (:4007)

Request flow, file by file

150 questions with answers

12 sections

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10 — notification-service (:4007)

Job: turn events into messages a human would want to read.

```
== Kafka ==
events/consumer.js  (same skeleton as §9: dedup, release, rethrow)
  subscribes: donation.assigned · accepted · unassigned
    ▼
events/handlers.js
  | domain/templates.js
  |   ▶ PURE function: event → { title, body, priority }
  |   ▶ priority HIGH on the ones that would justify waking a
  |     phone – carried, but nothing delivers push yet
  |
  |─ models/Notification.js  insert
    ▶ unique PARTIAL index on (sourceEventId, recipientId)
    makes a duplicate physically IMPOSSIBLE even if the
    Redis dedup let one through – a second line of defence
    ▶ partial, because not every notification comes from an
    event; without the filter every null would collide

== HTTP ==
routes/notification.routes.js      requireAuth
  | GET  /unread-count
  | GET  /                               → controllers/notification.controller.js
  | POST /read-all                     ↳ models/Notification.js
  |─ PATCH /:id/read
    ▶ scoped to req.user.id – you can only read your own
```

Appendix — the shared skeleton

Repeated here so this page stands alone. Every Node service has these same files doing the same jobs.

Every Node service has these same files doing the same jobs. Once you know this, each service below is just "the skeleton, plus X".

```
src/
├─ index.js           ▶ bootstrap: connect dependencies, THEN listen
├─ app.js             ▶ build the middleware chain (no port binding)
├─ config/
│   ├─ env.js         ▶ read + validate env vars, fail fast at startup
│   ├─ db.js          ▶ mongoose connection
│   └─ redis.js       ▶ redis client (only where Redis is used)
├─ middleware/
│   ├─ requestContext.js ▶ x-trace-id in/out, attaches req.log
│   ├─ requireAuth.js   ▶ verify JWT → req.user; requireRole(...) gate
│   ├─ validate.js      ▶ schema check → 400 with field-level details
│   └─ errorHandler.js ▶ the ONE place every error is formatted
├─ routes/            *.routes.js ▶ URL → middleware chain → controller
│                      *.schemas.js ▶ the validation schemas
├─ controllers/       ▶ the actual work for one endpoint
├─ models/            ▶ mongoose schemas
└─ utils/
    ├─ jwt.js          ▶ signToken / verifyToken
    ├─ ApiError.js     ▶ typed errors (badRequest, notFound, forbidden...)
    └─ logger.js       ▶ structured JSON logging
```

The universal request pipeline



Why `index.js` and `app.js` are separate. `app.js` builds the Express app but never binds a port; `index.js` connects the database *first*, then listens. That split is what lets the tests drive the app in-process with `supertest` without binding a port or racing on one — and it means a service never accepts a request it cannot serve.

Why the error handler is last. Every failure, from any middleware or controller, funnels into one place. That is also where donation-service cleans up orphaned uploads — putting cleanup on the happy path missed every request that failed validation.